TLS Guide for Thaghrah

Description

TLS (Transport Layer Security) is the standard protocol used to provide **secure communication** over a computer network. It is the successor to the older SSL (Secure Sockets Layer) protocol and is widely used to encrypt connections, especially for HTTPS websites, email, messaging apps, and more.

TLS ensures three main security properties:

- **Confidentiality** — Data is encrypted so only the intended recipient can read it.
- **Integrity** — Data cannot be tampered with during transit.
- **Authentication** — Verifies that you are communicating with the legitimate server (via certificates).

TLS operates between the Transport Layer (TCP) and Application Layer, adding a security layer to otherwise insecure protocols like HTTP, SMTP, and FTP.

Key Components of TLS

1. TLS Handshake

This is the most important process:

- Client and server negotiate the TLS version and cipher suite.
- Server presents its digital certificate for authentication.
- They exchange keys to establish a shared symmetric encryption key.
- Secure encrypted session begins.

2. Certificates and Public Key Infrastructure (PKI)

- Servers present **X.509 certificates** signed by trusted Certificate Authorities (CAs).
- Certificates contain the server's public key and identity.

3. Cipher Suites

Combinations of algorithms for:

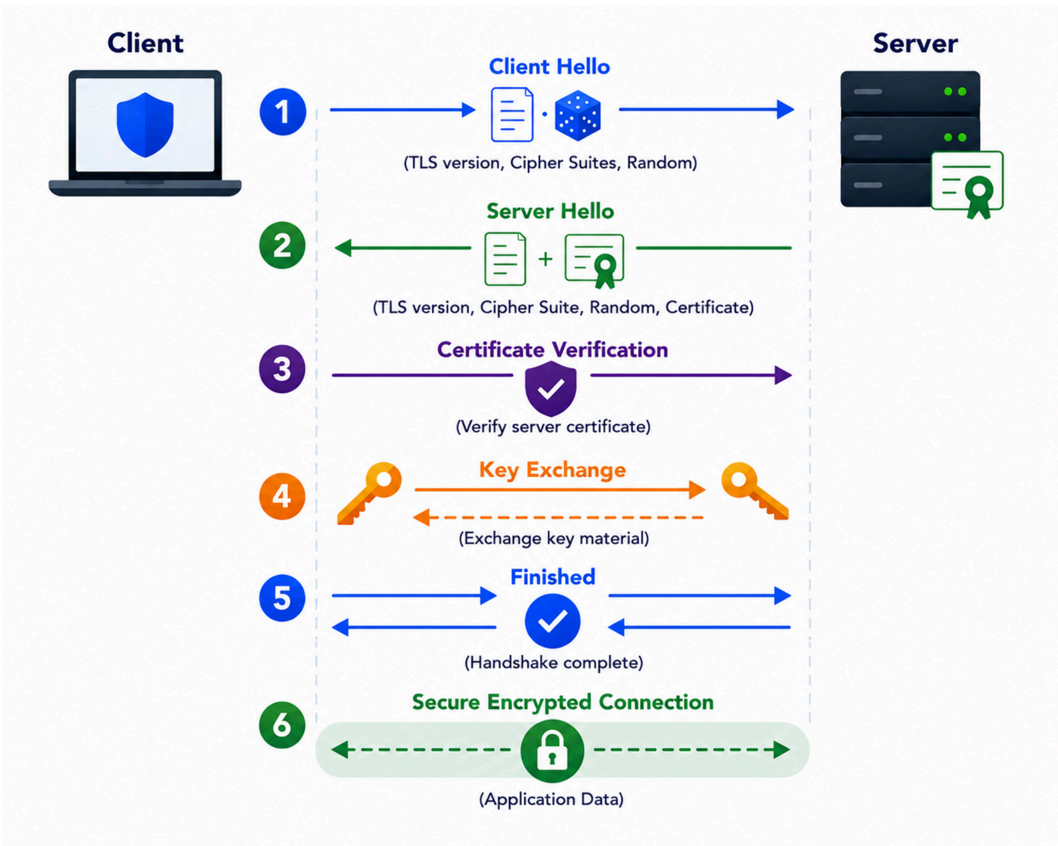
- Key exchange (e.g., ECDHE)
- Authentication
- Encryption (e.g., AES)
- Message Authentication (e.g., SHA-256)

4. TLS Record Protocol

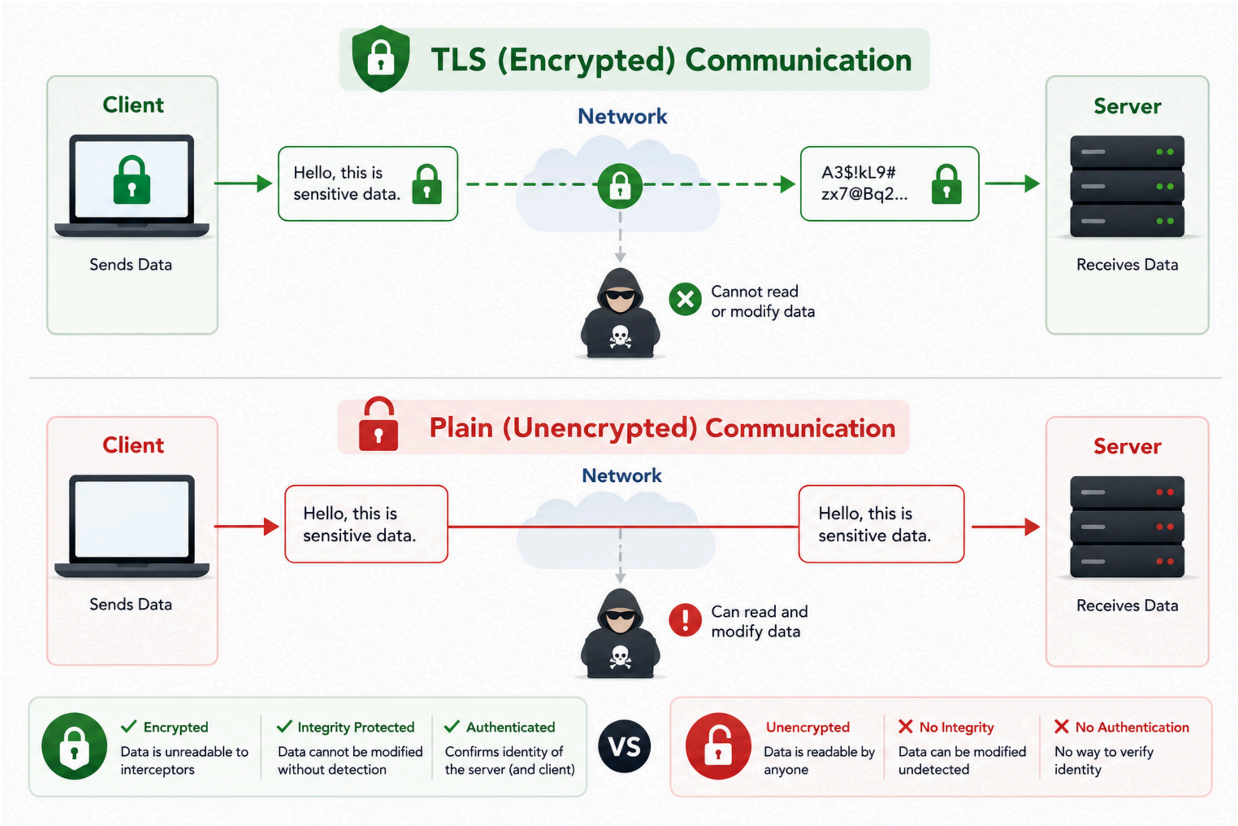
After the handshake, all application data is broken into records, encrypted, and sent with integrity protection.

Visualizations

TLS Handshake Process:



TLS vs Plain Communication:



SSL/TLS Explained in 7 Minutes

https://youtu.be/67Kfsmy_frM?si=AEVhhPW4Wvo4u5f3